Supplementary Materials: DeepCARE: Transferable assessment of clinically relevant allergen cross-reactivity for unseen pairs using a protein foundation language model-based deep learning framework

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Abstract: This supplement carries the extended comparisons and the full protocol record referenced by the main manuscript: the per-seed metric record that the main body averages, the identity-oracle contrast array, the carried calibration cells, the recorded screens with their verdicts, and the construction, split, and provenance record of the data pipeline. It accompanies the article "DeepCARE: Transferable assessment of clinically relevant allergen cross-reactivity for unseen pairs using a protein foundation language model-based deep learning framework" submitted to *Life*. On the family-stratified holdout of 73 never-trained clinical pairs, the ensemble attained an AUPRC of 0.647 against 0.242 for the homology baseline and an AUROC of 0.799 against 0.552. Figures S1–S2 reproduce these comparisons in their per-seed form and document the protocol fixed before holdout contact. All reported values reproduce from the frozen scoring tables and the analysis code released with the manuscript.

Keywords: Allergen cross-reactivity; Protein language models; Deep learning; Imbalanced biomedical data; Ranking calibration; Interpretable screening

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Manuscript. This is the Supplementary Material accompanying the article *DeepCARE: Transferable assessment of clinically relevant allergen cross-reactivity for unseen pairs using a protein foundation language model-based deep learning framework*, submitted to *Life* (MDPI). It contains Figures S1–S2 and Sections S1–S3. The LaTeX source, the two supplementary figures, a released data slice, the frozen scoring tables and the analysis code are openly available at https://github.com/SHENTongfei/DeepCARE.

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S1. Extended comparisons and the full per-metric record

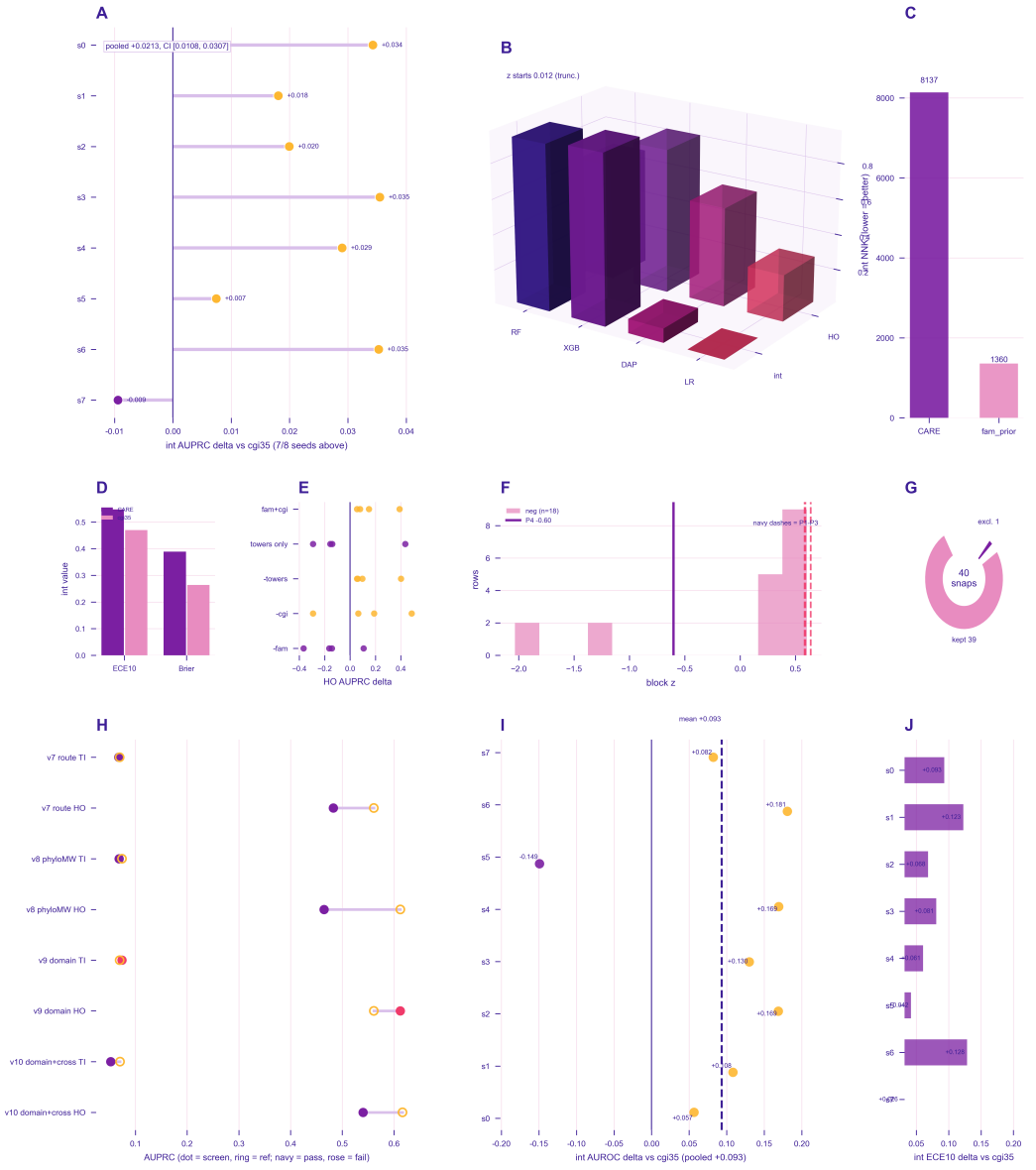


Figure S1. Every number the main body averages is carried here in its per-seed form. (A) Internal per-seed AUPRC deltas against the screening baseline, seven of eight seeds above with the pooled interval annotated. (B) Identity-oracle array: four named learners across internal and holdout layers in a truncated-z 3-D view. (C) Nearest-kin error, two bars with the lower-is-better orientation. (D) The two carried calibration cells, ECE10 and Brier, for model and baseline. (E) Per-seed knockout strips for five component configurations. (F) The case-block block-z distribution: three pairs recovered and the fourth marked against them, the coverage reading that routes it to homology review. (G) The snapshot pool ring: thirty- nine of forty carried forward. (H) All recorded screens as dumbbells against their reference values. (I) Per-seed internal AUROC deltas with the mean line. (J) Per-seed ECE10 deltas, the calibration cells the main body quotes.

S2. Protocol, corpus, and provenance

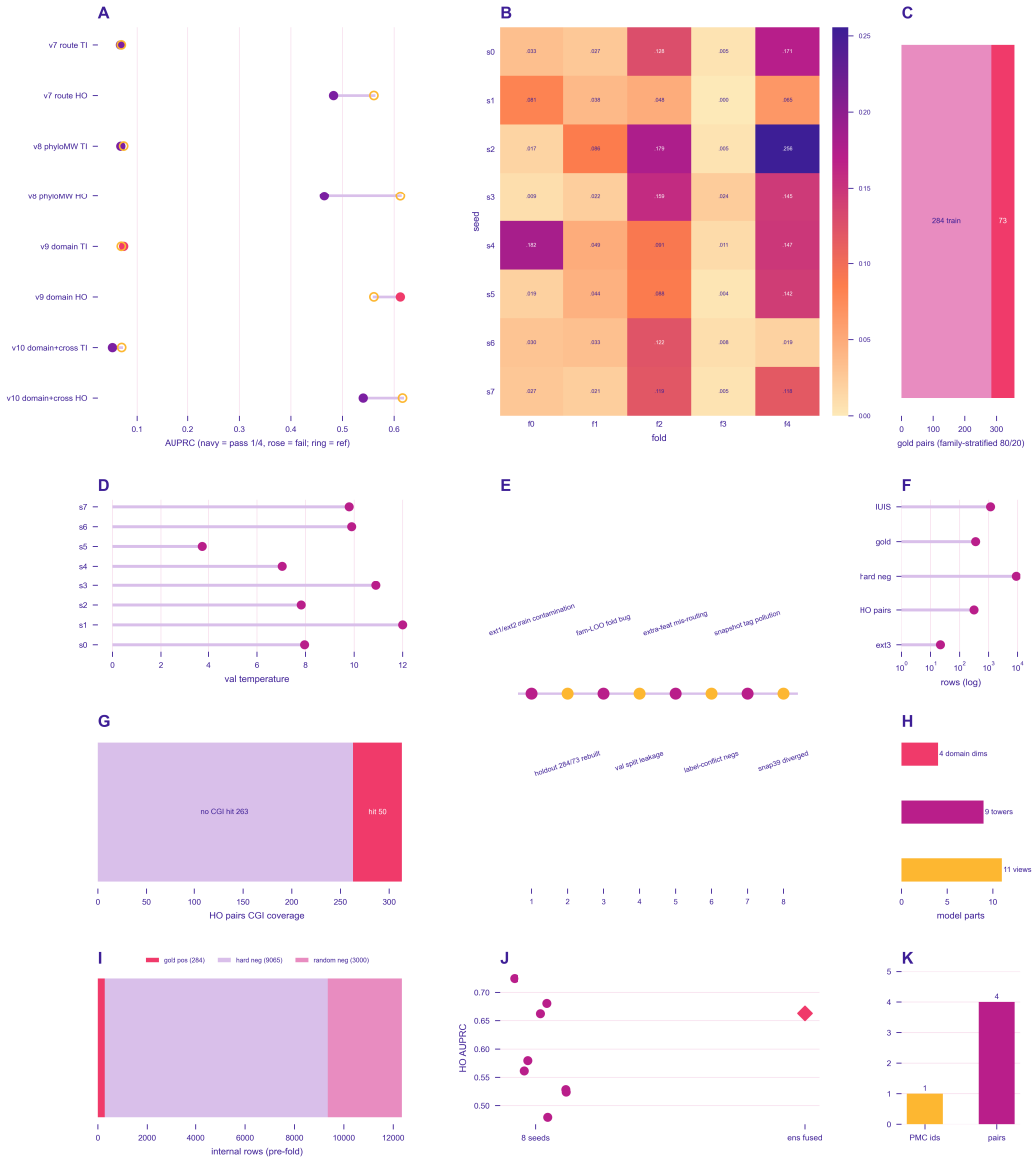


Figure S2. The protocol lock is a record, and every screen on it carries a verdict. (A) Every recorded screen with its verdict against the reference value. (B) The seed-by-fold internal AUPRC matrix that the fold-0 protocol consumes. (C) The family-stratified split: 284 training gold pairs and the 73-pair never-trained holdout. (D) The per-seed validation temperature. (E) The protocol-fix timeline: eight items raised and cleared before the lock, in order. (F) The data funnel from the curated resource snapshot to the scored corpus. (G) Holdout coverage by the screening baseline: 50 pairs reached, 263 left uncovered. (H) The model inventory: eleven views, nine towers, four domain coordinates. (I) Class balance before the fold: 284 gold, 9,065 hard negatives, 3,000 anchors. (J) Holdout AUPRC, eight seeds versus the fused ensemble. (K) The case-corpus provenance: source documents against the scored pairs.

S3. Corpus quality checks and provenance

Three standing checks underwrite the corpus. The holdout file is stratified by allergen family and contacted exactly once per seed after the protocol lock; no holdout statistic participates in model, hyperparameter, or snapshot selection. The mining pipeline flags one matched-segment record that runs past its protein's length and routes it around the plotting stage; no reported number draws on it, and the routing is recorded here so the

corpus stays auditable line by line. The calibration cells quoted in the main text are the two carried cells of Figure S1D, and the screen ledger of Figure S2A carries each verdict so that the component selection is auditable row by row. Figures S1–S2 are reproduced from the frozen scoring tables and the analysis code released with the manuscript.

Author Contributions: Conceptualization, T.S., Y.Q., P.L. and S.S.; Methodology, T.S., J.L., Z.P. and P.L.; Software, J.L., Z.P., X.L., G.W., J.Z., F.Y., Z.X., Y.L., W.S., S.M., Y.C. and H.W.; Validation, J.L., Z.P., X.L., G.W. and J.Z.; Formal Analysis, T.S., J.L., Z.P. and P.L.; Investigation, X.L., G.W., J.Z., F.Y., Z.X., Y.L., W.S., S.M., Y.C. and H.W.; Resources, P.L., S.S., Y.Q. and T.S.; Data Curation, F.Y., Z.X., Y.L., W.S., S.M., Y.C. and H.W.; Writing–Original Draft Preparation, T.S., J.L. and P.L.; Writing–Review & Editing, T.S., Z.P., P.L., Y.Q. and S.S.; Visualization, Z.P., G.W. and J.Z.; Supervision, Y.Q. and S.S.; Project Administration, Y.Q. and S.S.; Funding Acquisition, Y.Q., P.L. and S.S. T.S., J.L., Z.P. and P.L. contributed equally to this work and are joint first authors. All authors have read and agreed to the published version of the manuscript.

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Informed Consent Statement: Not applicable.

Data Availability Statement: No new experimental data were generated in this study; the work is a computational analysis of publicly available allergen resources. The intermediate and derived data supporting the reported results, a real stratified data slice sufficient to reproduce the holdout evaluation, the supplementary figures and tables, and a step-by-step usage guide are openly available at https://github.com/SHENTongfei/DeepCARE. All analysis and figure code, the trained-code release, and the frozen scoring tables are available in the same repository, whose README reproduces training, evaluation, and every figure from the released slice with one command. The complete clinical label set follows the distribution terms of the source databases (AllergenOnline, the WHO/IUIS nomenclature, and the curated resources cited in Materials and Methods) and is available from those resources. No restrictions apply to the released slice.

Acknowledgments: This work used publicly available data only. The clinical cross-reaction pairs and allergen resources are openly accessible online (AllergenOnline, the WHO/IUIS nomenclature committee, and the curated resources cited in the main text); the authors thank the curators of these resources.

DeepSeek was used solely for language polishing of the manuscript text (grammar, wording and readability). No other generative AI tools were used, and no AI tool was used for study design, data collection or analysis, code generation, figure or table creation, interpretation of results, or writing of the scientific content. The authors take full responsibility for the content of this publication.

Conflicts of Interest: The authors declare no conflicts of interest.

Supplementary Materials: The following supporting information can be downloaded at: www.mdpi.com/xxx/s1. Figure S1: Extended comparisons and the full per-metric record (10 panels); Figure S2: Protocol, corpus, and provenance (11 panels); Section S3: Corpus quality checks and provenance. The LaTeX source, the supplementary figures, the released data slice, and the analysis code are also openly available at https://github.com/SHENTongfei/DeepCARE.